7th AIP Early-Career Conference for Historians of the Physical Sciences

Name: Shivam Gupta

E-mail address: shivam1720406@gmail.com

Institutional affiliation: Independent Scholar

Presentation title: What Should the Acid Leave Behind? Resin, Soil, and the Repair of Fallout Measurements, 1960–1961

Abstract

Why could leaving radioactive material undissolved improve one fallout measurement and compromise another? In 1961, the US Atomic Energy Commission's Health and Safety Laboratory replaced destructive processing of contaminated collection resin with acid extraction, intended to leave unwanted strontium-90 inside the beads. In the same report, incomplete acid extraction from soil remained a reason to question, but not yet correct, accumulated-fallout estimates. Reading these episodes together with contemporary accounts of quality control, I examine how laboratories decided which activity belonged in an environmental number. A reproducible reanalysis of all sixteen published soil comparisons preserves a table-summary discrepancy and tests the limits of a proposed universal correction. A separate calculation shows why another apparent discrepancy can be explained by rounding. Unused-material checks supported a resin correction; contaminated fusion facilities and missing replacement replicates impeded a soil correction. The comparison locates environmental knowledge in selective dissolution, material provenance, and the institutional capacity to repeat a test. It offers a history of how measurements were repaired, beyond the expansion of monitoring networks.

Short biography

Shivam Gupta is an early-career independent scholar with a BTech in Computer Science and Design from Indraprastha Institute of Information Technology Delhi. He works in applied AI, software engineering, and product development, and is the founder of Siloed. His current historical research examines the material and computational practices through which scientific records become comparable and credible. This study combines close reading of published radiochemical reports with a source-linked transcription and reproducible numerical analysis. He does not hold a PhD.